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## **Discussion: Risk and Return in Asset Demand Systems**

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## A Strong Improvement on the Baseline Koijen-Yogo Model

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- This paper shows how to relax the multinomial logit framework from Koijen & Yogo (2019).
  - KY assumes constant risk/return tradeoff  $\frac{\Gamma_{it}(n)}{\mu_{it}(n)}$  across all assets
  - This paper shows how to solve the econometric difficulties that come from relaxing this assumption and demonstrates that doing so overcomes some of the theoretical and practical limitations of the IIA assumption.
- Two Major Contributions Beyond the Econometrics:
  - Alpha recovery for different categories of investors.
  - Low own-price elasticities are not mechanical outcomes of logit asset pricing models (Proposition 4).

## Why This Paper is Necessary

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- The past several years have seen various criticisms of the original KY paper as well as demand based asset pricing broadly.
  - Logit imposes a structural assumption that all assets are either uniformly substitutes or uniformly complements. Example given in the paper is Tesla & Panasonic vs Tesla & Ford. This is a major criticism from Fuchs et al (2025).
  - Estimated own-price elasticities from logit asset pricing models are wildly different from the calibrated CAPM literature. (Petajisto 2009)
- This paper is one of several new working papers grappling seriously with these criticisms

# From mean-variance portfolio choice to logit asset demand (KY 2019)

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Investor  $i$  at time  $t$  chooses portfolio weights  $\mathbf{w}_{i,t}$  over  $N_{i,t}$  assets.

**Starting point: mean-variance portfolio weights**

$$\mathbf{w}_{i,t} \approx \Sigma_{i,t}^{-1} (\mu_{i,t}) \quad (\Sigma_{i,t} \text{ has } N(N+1)/2 \text{ free parameters})$$

**Restriction 1.** Factor return structure  $\Sigma_{i,t} = \Gamma_{i,t} \Gamma'_{i,t} + \gamma_{i,t} \mathbf{I}$ .

$$\mathbf{w}_{i,t} = \frac{1}{\gamma_{i,t}} (\mu_{i,t} - \kappa_{i,t} \Gamma_{i,t}), \quad \kappa_{i,t} \equiv \mathbf{w}'_{i,t} \Gamma_{i,t} \quad (\text{portfolio factor exposure})$$

**Restriction 2.** Homogeneous risk-return tradeoff:  $\Gamma_{i,t}(n)/\mu_{i,t}(n) = c_{i,t}$  for all  $n$ . The common ratio cancels out from the relative weights, leaving:

$$\frac{w_{i,t}(n)}{w_{i,t}(m)} = \frac{\mu_{i,t}(n)}{\mu_{i,t}(m)} \quad (\text{Independence of Irrelevant Alternatives})$$

**Restriction 3.** Exponential expected return  $\mu_{i,t}(n) = \exp(\mathbf{x}'_t(n) \beta_{i,t}) \varepsilon_{i,t}(n)$ :

$$\frac{w_{i,t}(n)}{w_{i,t}(0)} = \exp(\mathbf{x}'_t(n) \beta_{i,t} + \beta_{K,i,t}) \varepsilon_{i,t}(n)$$

 (Kojien-Yogo)

*Substantive Takeaway:* Investors perceive all assets as uniformly substitutes or uniformly complements. The sign of  $\beta_{me,i,t}$  determines which.

## Akbas-Wang's relaxation: heterogeneous risk-return tradeoff

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**Drop only Restriction 2.** Let the perceived risk-return tradeoff depend on observable characteristics:

$$\frac{\Gamma_{i,t}(n)}{\mu_{i,t}(n)} = 1 + \mathbf{x}'_t(n) \Delta_{i,t} \quad (\text{new structural parameter } \Delta_{i,t})$$

**Resulting asset demand:**

$$\frac{w_{i,t}(n)}{w_{i,t}(0)} = \exp(\beta_{K,i,t}) \mu_{i,t}(n) \underbrace{\left[ 1 - \mathbf{x}'_t(n) \Delta_{i,t} \cdot \frac{\kappa_{i,t}}{1 - \kappa_{i,t}} \right]}_{\text{IIA broken via } \kappa_{i,t}}$$

Now the relative portfolio weight depends on  $\kappa_{i,t}$ , which aggregates over *all* assets in the portfolio. Relative demand for asset  $n$  vs  $m$  now depends on what else investor  $i$  holds. (Tesla & Panasonic example)

**Own-price elasticity decomposition:**

$$-\frac{\partial q_{i,t}(n)}{\partial p_t(j)} = \underbrace{\quad}_{\text{logit channel: via } \beta_{\text{price}}} \text{ Diversion} + \underbrace{\quad}_{\text{new: via } \Delta_{i,t}} \text{ Intra-portfolio} + \underbrace{\quad}_{\text{new: via } \partial \kappa_{i,t} / \partial p} \text{ Portfolio-level}$$

Within one investor's portfolio, some asset pairs can now be substitutes and others can be complements.

## What is the Economic Interpretation of $\Delta_{me}$ ?

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- In this model,  $\Delta_{i,t}$  is a parameter vector that allows the risk-return tradeoff  $\frac{\Gamma_{it}(n)}{\mu_{it}(n)}$  to vary in observed characteristics  $\mathbf{x}_t(n)$
- By construction,  $\Delta_{i,t,me} < 0$  means small stocks have a worse risk-return tradeoff than large stocks.
  - The paper finds  $\Delta_{me} = -0.42$  and that it is significantly negative for 55% of all (investor x quarter) observations.
  - This is the opposite of the SMB anomaly documented in Benz (1981), Fama-French (1992,1993), and Asness et al (2018).
- How are we to interpret the economic meaning of this strongly negative relationship in the model and how do we link it to the existing literature?
  - Are investors over-valuing the perceived stability of megacap firms especially post 2010?
  - Does this estimated  $\Delta_{me}$  change across time?

## Are "Complements Investors" Really such?

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- This paper estimates a larger own-price elasticity than the logit benchmark model does and attributes it to "complements investors", or investors with  $\beta_{me}^{MNL} \in (0, 1)$ .
- However, most of these investors are ones structurally required to own the whole market or close to it (passive funds, brokers, long term pension funds).
- Are these investors really being driven by viewing different stocks as complements? (the Tesla and Panasonic example)
  - Or, does this just reflect mechanical passive investing rules?
- The economic interpretation "Investors view Tesla and Panasonic as complements because their businesses are synergistic" is quite different from "Investors hold Tesla and Panasonic together because they just buy everything on the market".

## The gap between elasticity estimates

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- The paper proves that logit demand systems can generate arbitrarily high own price elasticities. However the modeled  $\beta_{me}$  in this paper is still very small compared to calibrated CAPM estimates, leading to the fundamental disconnect between the two literatures remaining.
  - Davis-Kargar-Li (2025) claim that classical models overstate the price elasticity by assuming very low 'unspanned returns' (essentially assuming very high substitutability between assets).
  - Fuchs et al (2025) claim that if assets are highly substitutable then demand system elasticity estimates can be infinitely biased in the limit.
  - While proposition 4 is a necessary step forward towards resolving this tension the authors don't directly address either of these points.



## Some Smaller Issues

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- The factor benchmark  $f_{i,t+1}$  in the model's return structure is investor specific.
  - This complicates the comparison of alpha between investors since the alphas calculated are relative to different benchmark factors.
  - Do brokers actually outperform hedge funds or do hedge funds have a richer factor structure?
- In section 4 the log portfolio weight is approximated using a second order Taylor expansion for intuition.
  - However, the estimates of  $ME_{i,t} = 0.42$  are significantly away from zero. The intuition is local but the actual estimator uses global identification from the GMM.
  - In practice, BLP-related estimators often have problems with weak identification of the nonlinear terms even if the rank-condition for identification is met.
  - The intuition may obscure this fact and it would be nice to see empirical estimates of how strong the identification of these nonlinear terms actually is under the GMM estimator.

## Conclusion

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- This paper takes the model blueprint of Koijen & Yogo (2019) and relaxes its modeling assumptions away from logit.
  - It does this by allowing different investors to have different perceived risk-return tradeoffs.
  - One of several papers exploring alternative substitution patterns in demand based asset pricing (Chaudhary-Fu-Li (2025), Allen-Kastl-Wittwer (2026))
- It is an important step forward, but more clarity about the economic interpretation of the key findings would be desirable.
- I expect this segment of the literature to continue to grow in the near future.